

LOGIC OVERFLOW

Mastering Loops & Control Structures | Python & DSA Cohort

Assignment: Master-Level Nested Loop Patterns

Objective: Develop structural debugging skills, spatial matrix mapping, and loop iteration logic. Solve all 20 problems below using optimized nested `for` loops. Avoid using built-in string multiplication tricks (like `"*" * n`) to maximize logic building.

1. Solid Square Grid

Print a solid square grid of asterisks (`\*`) with dimensions N x N.

Sample Input Constraint	Expected Terminal Output
N = 4	**** **** **** ****

2. Right-Angled Triangle

Print a right-angled triangle pattern where row number dictates the count of asterisks.

Sample Input Constraint	Expected Terminal Output
N = 4	* ** *** ****

3. Inverted Right-Angled Triangle

Print a right-angled triangle upside down, decreasing the asterisk count per row.

Sample Input Constraint	Expected Terminal Output
N = 4	**** *** ** *

4. Mirrored Right-Angled Triangle

Print a right-angled triangle mirrored horizontally, using leading spaces.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre> * ** *** ****</pre>

5. Inverted Mirrored Right Triangle

Print an inverted right-angled triangle mirrored horizontally.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>**** *** ** *</pre>

6. Row-Number Pyramid

Print a right-angled triangle where each item displays its current row index value.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>1 22 333 4444</pre>

7. Continuous Count Progression

Print a right triangle using sequential numbers increasing systematically (Floyd's Triangle).

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>1 2 3 4 5 6 7 8 9 10</pre>

8. Binary Alternating Matrix

Print a right triangle of alternating 1s and 0s based on index positioning metrics.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>1 0 1 1 0 1 0 1 0 1</pre>

9. The Full Classic Pyramid

Print an equidistant space-padded classic odd-numbered asterisk pyramid.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre> * *** ***** *****</pre>

10. Inverted Classic Pyramid

Print a classic odd-numbered asterisk pyramid upside down.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>***** ***** *** *</pre>

11. The Diamond Star Configuration

Combine a standard and inverted pyramid sequence to structure a perfectly symmetric diamond.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre> * *** ***** ***** *** *</pre>

12. Hollow Square Frame

Print an N x N square outline using asterisks for borders and empty spaces internally.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>**** * * * * * * ****</pre>

13. The Numeric Palindromic Tree

Print a symmetric space-padded pyramid where each row reads as a numeric palindrome.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre> 1 121 12321 1234321</pre>

14. Alphabet Character Escalation

Print a right-angled triangle changing letters chronologically by row positioning index.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>A BB CCC DDDD</pre>

15. Continuous Alphabet Flow

Print a sequential right triangle where character variables increment across lines consistently.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>A B C D E F G H I J</pre>

16. Horizontal Double Triangle (Arrow)

Print a right-pointing arrow structure by running an increasing triangle into a decreasing one.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>* ** *** **** *** ** *</pre>

17. Hollow Pyramid Structure

Print a standard odd pyramid outline, preserving only the boundary asterisks.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre> * * * * * *****</pre>

18. The Cross Pattern Symbol (X)

Print an N x N dimensional matrix representing an 'X' symbol framework using diagonal detection.

Sample Input Constraint	Expected Terminal Output
N = 5	<pre>* * * * * * * * *</pre>

19. Hourglass Matrix Layout

Merge an inverted pyramid structure directly into a standard odd pyramid layout.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre>***** ***** *** * *** ***** *****</pre>

20. Pascal's Triangulation

Print an arithmetic triangle pattern calculating adjacent positional item coefficients.

Sample Input Constraint	Expected Terminal Output
N = 4	<pre> 1 1 1 1 2 1 1 3 3 1</pre>